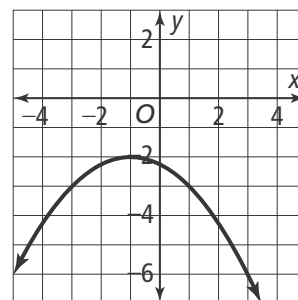


1-2 Lesson Quiz

Transformations of Functions

1. Select all the transformations of $f(x) = x^2$ that combine to result in the graph of function g at the right.

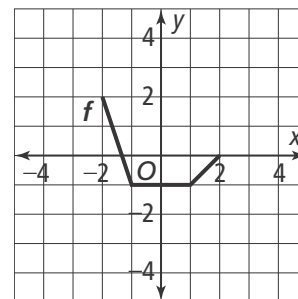
- ☐ A. translation of 1 unit left
- ☐ B. translation of 1 unit right
- ☐ C. translation of 2 units down
- ☐ D. horizontal stretch by a factor of 2
- ☐ E. vertical stretch by a factor of 2
- ☐ F. reflection across the x -axis



2. What is the equation for g , which is $f(x) = 2x^2 + 3x - 1$ reflected across the y -axis?

- ☐ A $g(x) = 2x^2 + 3x - 1$
- ☐ B $g(x) = -2x^2 - 3x + 1$
- ☐ C $g(x) = 2x^2 - 3x - 1$
- ☐ D $g(x) = -2x^2 - 3x - 1$

3. The graph represents f . Graph $g(x) = 2f(x - 1)$.



4. A bucket has a leak that started 6 hours ago. At time 0, when the bucket just emptied, a tap was turned on for 3 hours. The amount of water in the bucket can be modeled by the function $f(x) = |x|$ over the interval $[-6, 3]$, where x is the time, in hours, and y is the amount of water in the bucket, in gallons. A similarly full bucket started to leak only 3 hours ago. Which translation of the graph represents the amount of water in the second bucket if it is moved under the tap as soon as it empties?

- ☐ A $g(x) = |x| + 3$
- ☐ B $g(x) = |x + 3|$
- ☐ C $g(x) = |x| - 3$
- ☐ D $g(x) = |x - 3|$

5. The graph of the function f is shown. The domain of f is $[0, 50]$. What is the range of $g(x) = 4f(x)$?

- ☐ A $[0, 50]$
- ☐ B $[0, 70]$
- ☐ C $[0, 1.75]$
- ☐ D $[0, 7]$

